# Practice Final Exam Solutions

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## Exercise 1: Complex Numbers & Roots

(a) Find all complex roots of the equation $z^4 = -8 + 8i\sqrt{3}$. Express your answers in exponential form $re^{i\theta}$.

(b) Let $w = \dfrac{2-i}{3+i}$. Calculate its modulus $|w|$, its complex conjugate $\overline{w}$, and express $w$ in the standard Cartesian form $x + iy$.

**Solution:**

**(a)** Write $-8 + 8i\sqrt{3}$ in polar form. Magnitude $r = \sqrt{64 + 64(3)} = \sqrt{256} = 16$. The point is in the second quadrant, so $\theta = \pi - \pi/3 = 2\pi/3$. Thus, $-8 + 8i\sqrt{3} = 16e^{i2\pi/3}$. The roots are $z_k = \sqrt[4]{16}e^{i(2\pi/3+2k\pi)/4} = 2e^{i(\pi/6+k\pi/2)}$ for $k = 0, 1, 2, 3$.

$$z_0 = 2e^{i\pi/6}, \quad z_1 = 2e^{i2\pi/3}, \quad z_2 = 2e^{i7\pi/6}, \quad z_3 = 2e^{i5\pi/3}$$

**(b)** Standard Cartesian form:

$$w = \frac{2-i}{3+i} \cdot \frac{3-i}{3-i} = \frac{6-2i-3i-1}{3^2+1^2} = \frac{5-5i}{10} = \frac{1}{2} - \frac{1}{2}i$$

Modulus: $|w| = \sqrt{(1/2)^2 + (-1/2)^2} = \sqrt{2/4} = \frac{1}{\sqrt{2}}$.
Conjugate: $\overline{w} = \frac{1}{2} + \frac{1}{2}i$.

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## Exercise 2: Analytic Functions & Harmonic Conjugates

Consider the real-valued function $u(x, y) = e^x \cos(y) - xy$.

(a) Verify that $u(x, y)$ is harmonic in the entire complex plane.

(b) Find the harmonic conjugate $v(x, y)$ such that $f(z) = u(x, y) + iv(x, y)$ is an analytic function.

(c) Express $f(z)$ explicitly in terms of the complex variable $z$.

**Solution:**

**(a)** Check $\Delta u = u_{xx} + u_{yy} = 0$:

$$u_x = e^x \cos y - y \implies u_{xx} = e^x \cos y$$

$$u_y = -e^x \sin y - x \implies u_{yy} = -e^x \cos y$$

$u_{xx} + u_{yy} = e^x \cos y - e^x \cos y = 0$. Thus, $u$ is harmonic.

**(b)** Using Cauchy-Riemann: $v_y = u_x = e^x \cos y - y$. Integrate w.r.t $y$: $v(x,y) = e^x \sin y - \frac{y^2}{2} + h(x)$. Differentiate w.r.t $x$: $v_x = e^x \sin y + h'(x)$. By the second CR equation, $v_x = -u_y = e^x \sin y + x$. Equating the two: $e^x \sin y + h'(x) = e^x \sin y + x \implies h'(x) = x \implies h(x) = \frac{x^2}{2} + C$.

$$v(x,y) = e^x \sin y + \frac{x^2 - y^2}{2}$$

**(c)** Express $f(z) = u + iv$:

$$f(z) = (e^x \cos y - xy) + i(e^x \sin y + \frac{x^2 - y^2}{2})$$

$$f(z) = e^x(\cos y + i \sin y) + i\frac{x^2 - y^2 + 2ixy}{2} = e^z + i\frac{z^2}{2}$$

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## Exercise 3: Complex Integration

Use the Generalized Cauchy Integral Formula to evaluate the following contour integral:

$$\oint_{|z|=2} \frac{ze^{\pi z}}{(z-i)^3}\, dz$$

where the contour $|z| = 2$ is traversed in the counterclockwise direction.

**Solution:**

The integrand has a pole of order 3 at $z_0 = i$, which lies inside $|z| = 2$. By Generalized CIF:

$$\oint_{|z|=2} \frac{ze^{\pi z}}{(z-i)^3}\, dz = \frac{2\pi i}{2!} \frac{d^2}{dz^2}[ze^{\pi z}]\bigg|_{z=i}$$

Let $g(z) = ze^{\pi z}$.

$$g'(z) = e^{\pi z} + \pi z e^{\pi z}$$

$$g''(z) = \pi e^{\pi z} + \pi e^{\pi z} + \pi^2 z e^{\pi z} = e^{\pi z}(2\pi + \pi^2 z)$$

Evaluate at $z = i$:

$$g''(i) = e^{i\pi}(2\pi + i\pi^2) = -1(2\pi + i\pi^2) = -2\pi - i\pi^2$$

Finally:

$$\text{Integral} = \pi i(-2\pi - i\pi^2) = -2\pi^2 i - i^2\pi^3 = \pi^3 - 2\pi^2 i$$

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## Exercise 4: Laurent Series

Consider the rational function:

$$f(z) = \frac{5z}{(z-1)(z+4)}$$

(a) Find the partial fraction decomposition of $f(z)$.

(b) Compute the Laurent series expansion for $f(z)$ centered at $z_0 = 0$ strictly valid in the disk $|z| < 1$.

(c) Compute the Laurent series expansion for $f(z)$ centered at $z_0 = 0$ strictly valid in the annulus $1 < |z| < 4$.

(d) Compute the Laurent series expansion for $f(z)$ centered at $z_0 = 0$ strictly valid in the region $|z| > 4$.

**Solution:**

**(a) Partial Fraction Decomposition:**

$$\frac{5z}{(z-1)(z+4)} = \frac{A}{z-1} + \frac{B}{z+4} \implies A(z+4) + B(z-1) = 5z$$

Let $z = 1 \implies 5A = 5 \implies A = 1$. Let $z = -4 \implies -5B = -20 \implies B = 4$.

$$f(z) = \frac{1}{z-1} + \frac{4}{z+4}$$

**(b) Disk $|z| < 1$:** Both $|z| < 1$ and $|z/4| < 1$.

$$f(z) = -\frac{1}{1-z} + \frac{1}{1-(-z/4)} = -\sum_{n=0}^{\infty} z^n + \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{4^n} = \sum_{n=0}^{\infty} \left[ \frac{(-1)^n}{4^n} - 1 \right] z^n$$

**(c) Annulus $1 < |z| < 4$:** Here $|1/z| < 1$ and $|z/4| < 1$.

$$\frac{1}{z-1} = \frac{1}{z(1-1/z)} = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}}$$

$$f(z) = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} + \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{4^n}$$

**(d) Region $|z| > 4$:** Both $|1/z| < 1$ and $|4/z| < 1$.

$$\frac{4}{z+4} = \frac{4}{z(1-(-4/z))} = \frac{4}{z} \sum_{n=0}^{\infty} (-1)^n \left( \frac{4}{z} \right)^n = \sum_{n=0}^{\infty} (-1)^n \frac{4^{n+1}}{z^{n+1}}$$

$$f(z) = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}} + \sum_{n=0}^{\infty} (-1)^n \frac{4^{n+1}}{z^{n+1}} = \sum_{n=0}^{\infty} \frac{1 + (-1)^n 4^{n+1}}{z^{n+1}}$$

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## Exercise 5: Real Improper Integrals

Use the Residue Theorem and a semicircular contour in the upper half-plane to evaluate the following improper real integral:

$$\int_{-\infty}^{\infty} \frac{x^2}{(x^2+4)^2} \, dx$$

**Solution:**

Let $f(z) = \frac{z^2}{(z^2+4)^2}$. We integrate over a semicircular contour $C$ in the upper half-plane. The roots of $(z^2+4)^2 = 0$ are $\pm 2i$, both of order 2. Only $z = 2i$ is in the UHP.

$$\text{Res}(f, 2i) = \lim_{z \to 2i} \frac{d}{dz} \left[ \frac{z^2}{(z+2i)^2} \right]$$

$$\frac{d}{dz}\left[\frac{z^2}{(z+2i)^2}\right] = \frac{2z(z+2i)^2 - z^2(2)(z+2i)}{(z+2i)^4} = \frac{2z(z+2i) - 2z^2}{(z+2i)^3} = \frac{4iz}{(z+2i)^3}$$

Evaluate at $z = 2i$:

$$\text{Res}(f, 2i) = \frac{4i(2i)}{(4i)^3} = \frac{-8}{-64i} = \frac{1}{8i} = -\frac{i}{8}$$

By the Residue Theorem:

$$\oint_C f(z)\, dz = 2\pi i \left(-\frac{i}{8}\right) = \frac{2\pi}{8} = \frac{\pi}{4}$$

The integral over $C_R$ vanishes because $|f(z)| \approx 1/R^2$. Thus, the real integral is $\pi/4$.

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## Exercise 6: Trigonometric Integrals

Use the Residue Theorem to evaluate the following definite integral:

$$\int_0^{2\pi} \frac{\cos(2\theta)}{5 - 4\cos(\theta)}\, d\theta$$

**Solution:**

Let $z = e^{i\theta} \implies d\theta = \frac{dz}{iz}$. Also, $\cos(2\theta) = \frac{z^2 + 1/z^2}{2}$ and $\cos\theta = \frac{z+1/z}{2}$.

$$\int_0^{2\pi} \frac{\cos(2\theta)}{5 - 4\cos(\theta)}\, d\theta = \oint_{|z|=1} \frac{\frac{z^4+1}{2z^2}}{5 - 2(z + z^{-1})} \frac{dz}{iz} = \frac{i}{2} \oint_{|z|=1} \frac{z^4+1}{z^2(2z^2 - 5z + 2)}\, dz$$

The roots of $2z^2 - 5z + 2 = (2z-1)(z-2)$ are $z = 1/2$ and $z = 2$. The poles inside $|z| = 1$ are $z = 0$ (order 2) and $z = 1/2$ (simple). Let $f(z) = \frac{z^4+1}{z^2(2z^2-5z+2)}$.

**Residue at $z = 1/2$:**

$$\text{Res}(f, 1/2) = \left.\frac{z^4+1}{z^2(2)(z-2)}\right|_{z=1/2} = \frac{(1/16)+1}{(1/4)(2)(1/2-2)} = \frac{17/16}{1/2(-3/2)} = \frac{17/16}{-3/4} = -\frac{17}{12}$$

**Residue at $z = 0$:**

$$\text{Res}(f, 0) = \lim_{z\to 0} \frac{d}{dz}\left[\frac{z^4+1}{2z^2 - 5z + 2}\right]$$

Using the quotient rule:

$$\frac{d}{dz}\cdots = \frac{4z^3(2z^2-5z+2) - (z^4+1)(4z-5)}{(2z^2-5z+2)^2}$$

Evaluate at $z = 0$:

$$\text{Res}(f, 0) = \frac{0 - (1)(-5)}{(2)^2} = \frac{5}{4}$$

**Final Evaluation:** Sum of residues $= \frac{5}{4} - \frac{17}{12} = \frac{15}{12} - \frac{17}{12} = -\frac{2}{12} = -\frac{1}{6}$. Integral $= \frac{i}{2}(2\pi i)\left(-\frac{1}{6}\right) = -\pi\left(-\frac{1}{6}\right) = \frac{\pi}{6}$.

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## Exercise 7: Conformal Mapping

Find the unique Möbius transformation $w = T(z)$ that maps the points:

$$z_1 = i, \quad z_2 = 1, \quad z_3 = -i$$

onto the points:

$$w_1 = -1, \quad w_2 = 0, \quad w_3 = 1$$

Simplify your final expression for $w$ as a function of $z$.

**Solution:**

Let the general Möbius transformation be given by:

$$w = T(z) = \frac{az+b}{cz+d}$$

We will substitute the three given points to find the constants $a, b, c$, and $d$.

**Step 1: Use the root.** We know $T(1) = 0$.

$$0 = \frac{a(1)+b}{c(1)+d} \implies a+b=0 \implies b=-a$$

Substitute this back into the general form to reduce the number of variables:

$$w = \frac{a(z-1)}{cz+d}$$

**Step 2: Substitute the remaining points.** Using $T(i) = -1$:

$$-1 = \frac{a(i-1)}{ci+d} \implies -ci-d = ai-a \quad \text{— (Equation 1)}$$

Using $T(-i) = 1$:

$$1 = \frac{a(-i-1)}{-ci+d} \implies -ci+d = -ai-a \quad \text{— (Equation 2)}$$

**Step 3: Solve the system for $c$ and $d$ in terms of $a$.** Add Equation 1 and Equation 2 together:

$$(-ci-d)+(-ci+d) = (ai-a)+(-ai-a)$$

$$-2ci = -2a \implies ci = a \implies c = \frac{a}{i} = -ai$$

Now, subtract Equation 1 from Equation 2:

$$(-ci+d)-(-ci-d) = (-ai-a)-(ai-a)$$

$$2d = -2ai \implies d = -ai$$

**Step 4: Build the final transformation.** We now have $b = -a$, $c = -ai$, and $d = -ai$. Substitute these into the original equation:

$$w = \frac{az-a}{-aiz-ai}$$

Factor out the common constant $a$ from the numerator and denominator:

$$w = \frac{a(z-1)}{a(-iz-i)} = \frac{z-1}{-i(z+1)}$$

To write this in standard form, multiply the numerator and denominator by $i$:

$$w = \frac{i(z-1)}{-i^2(z+1)} = \frac{i(z-1)}{z+1}$$

$$w = T(z) = i\frac{z-1}{z+1}$$